

Table 2. Review of spatial computational methods for cell-cell communication

Method	How it computes interaction	Statistics
cellchat	Law of mass action, multiplying the average expression values of all ligand/receptor subunits, cofactors, and complexes. For spatial context, the expression product is explicitly modulated by a continuous distance decay function	Permutation of cell labels
lianaP_morans	It employs global spatial bivariate autocorrelation statistic (Moran's I) across entire tissue. It also allows for local neighborhood-weighted cosine similarity to be calculated (not used here).	Permutation of gene expression vectors
mistyR	For each marker of interest, mistyR builds prediction models (linear model used in this benchmark but in general is algorithm-agnostic framework) that relate its expression to other markers in different spatial contexts.	mistyR retrieves feature importance as opposed to pvalue. The importances depend on prediction model used. For linear model feature importances are defined as t-values of the model coefficients calculated via Ordinary Least Squares (OLS)
NICHES	NICHES preserves single-cell resolution and computes spatial interaction for each pairs of single-cells within radius. For each edge, interaction score is calculated by cross-product of the normalized ligand and receptor expression in corresponding cells	Significance of interaction is determined using FindAllMarkers function from Seurat
spatialDM	SpatialDM identifies significant interactions by detecting spatially co-expressed ligand-receptor pairs through bivariate Moran's I (Moran's R).	Permutation by shuffling cell IDs
stlearn	Interaction scores are calculated for each spot based on expression of ligand and the expression of receptor in all surrounding spots.	stlearn uses 2-level permutation test. LR significance testing: generates a random background signal of non-interaction genes. P-values for each spot and LR pair are derived based on proportion of the background scores that had a score greater than the LRscore. Celltype specific testing: For specific spatial locations obtained from previous testing, stlearn permutes celltype information to assess if specific celltypes are over-represented.
seurat_wilcoxon	Non spatially aware method.	Wilcoxon test
cellphonedbv5	Computes interaction based on average expression across celltypes defined in the microenvironment file provided by the user.	Permutation of cell labels